RAMYA AKULA, PH.D.

AI Research Scientist | Trustworthy AI · NLP · Computational Biology

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PROFESSIONAL SUMMARY

Ph.D.-trained AI researcher and educator with 10+ years spanning interpretable machine learning, natural language processing, computational biology, academic research, teaching, and international engineering. Developed peer-reviewed methods in interpretable AI and semantic evaluation, including work associated with U.S. Patent 12,210,833 and DARPA Forward Riser recognition. Recent postdoctoral research applies reproducible single-cell transcriptomic analysis to lung aging and disease. Earlier industry and research experience spans aerospace engineering, medical-device software quality, AI research, urban-data systems, and healthcare software.

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar - Computational Biology & Machine Learning 2024-2025

The Ohio State University, Department of Internal Medicine (Pulmonary) | Columbus, OH, USA

- Developed reproducible pipelines to integrate multi-cohort single-cell RNA-sequencing datasets, including quality control, Harmony-based batch correction, cell-type annotation, differential expression, and pathway-level interpretation.
- Analyzed transcriptomic signatures of lung aging, cellular senescence, apoptosis, donor type, and chronic lung injury across human and mouse cohorts.
- Partnered with pulmonary scientists to translate biological questions into analytical plans, interpretable visualizations, and publication-ready findings; contributed to a 2026 peer-reviewed publication in The Journals of Gerontology.

Lecturer - Computer Science & M.S. FinTech 2023-2024

University of Central Florida | Orlando, FL, USA

- Designed and taught graduate and undergraduate courses in artificial intelligence, NLP, machine learning, data science, algorithms, blockchain, smart contracts, and programming.
- Mentored and advised approximately 15 undergraduate students and supervised four senior projects in computer science.

Graduate Research Assistant - Trustworthy AI & NLP 2018-2022

University of Central Florida, Complex Adaptive Systems Laboratory | Orlando, FL, USA

- Developed an interpretable multi-head self-attention architecture for sarcasm detection; the work received a U.S. patent and DARPA Forward Riser recognition.
- Proposed SPEED Score, a sentence-embedding-based metric for evaluating abstractive and extractive summarization.
- Researched explainable AI, figurative-language understanding, toxic-language detection, semantic evaluation, and AI for social good under DARPA-supported programs.
- Served as NSF I-Corps Entrepreneurial Lead, conducting customer discovery and commercialization activities for an AI-assisted qualitative-analysis platform.

Project Engineer 2015-2017

P3 Group GmbH | Hamburg, Germany

- Developed OSCAR (Offer-ability Specification Card Reader) for Airbus aircraft models.
- Worked on Curved Stiffened Panel Sizing (CSPS) data extraction, including analysis of aircraft-part stress calculations, data extraction and visualization, and automated modification-closure reports for A350 ESI configuration management.
- Developed a Java SE tool to convert ASCII data into user-readable PDF reports.

Master Thesis - Intelligent Visualization of Complex Social Networks 2014-2015

Technical University of Kaiserslautern | Kaiserslautern, Germany

- Implemented a visualization tool using hierarchical edge bundling and K-nearest neighbors to classify similar users, reduce visual clutter, and improve bundling strength in complex social networks.

Software Quality Assurance Analyst 2013-2014

EXCO GmbH | Mannheim, Germany

- Worked on ACCU-CHEK software for reading blood-glucose measurements from a medical device; developed and assessed software test cases.

Graduate Research Assistant 2013-2014

Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI) GmbH | Kaiserslautern, Germany

- Worked on 3D reconstruction for smart-society research: processed distorted images, calculated intrinsic and extrinsic camera parameters, and used camera calibration to estimate object rotation and displacement in the real world.

Software Engineer 2011-2013

Regional Center for Urban and Environmental Studies (RCUES) | Hyderabad, India

- Developed widgets and responsive web applications for visualization of urban-planning statistics across desktop and mobile environments.
- Developed a user-management system and a unified access point for datasets gathered from governmental organizations and ministries to improve administrative data retrieval.

Software Developer **2010-2011**

Inventurus Software Solutions | Hyderabad, India

- Developed a Java embedded-database application for health-insurance data management and optimized dynamic search across text and scanned documents using pattern matching and feature extraction.

Bachelor Thesis - Chat Client **2009-2010**

Satya Technologies | Hyderabad, India

- Implemented a Java Applet chat client with multithreaded message processing and a chat-export feature for maintaining local conversation history.

SELECTED RESEARCH, INNOVATION & RECOGNITION

- U.S. Patent 12,210,833 (2025): Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection.
- DARPA Forward Riser (2022): Recognition associated with research in interpretable machine learning for figurative-language understanding.
- NSF I-Corps Entrepreneurial Lead (2019): Customer discovery and commercialization activities for an AI-assisted qualitative-analysis platform.
- First Place, Startup Weekend Orlando (2021): Contributed to an AI-powered online youth-safety product concept.
- Research coverage listed in the prior CV includes NBC News, IEEE Spectrum, Los Angeles Times, Defense One, ScienceDaily, and ABC Australia.

SELECTED PUBLICATIONS

- Akula, R., et al. “Transcriptomic shifts in human lungs donated after brain and circulatory death and severe Obliterative Bronchiolitis.” The Journals of Gerontology, 2026.
- Akula, R., & Garibay, I. “SPEED Score: Sentence Pair Embeddings Based Evaluation Metric for Abstractive and Extractive Summarization.” LREC, 2022.
- Akula, R., & Garibay, I. “Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection in Social Media.” Entropy, 2021.
- Akula, R., & Garibay, I. “Explainable Detection of Sarcasm in Social Media.” ACL Workshop, 2021.

  Complete publication record on Google Scholar

TECHNICAL EXPERTISE

Natural Language Processing & Language Evaluation: Transformers, text summarization, semantic similarity, sentence embeddings, figurative-language understanding, toxicity detection, model evaluation.

Trustworthy AI & Machine Learning: Interpretability, explainable machine learning, deep learning, representation learning, transfer learning, statistical evaluation, human-AI collaboration.

Computational Biology: Single-cell RNA sequencing, Scanpy, Seurat, Harmony, differential expression, cell annotation, gene-set enrichment, transcriptomics.

Data & Scientific Computing: Python, R, PyTorch, TensorFlow, Jupyter, Git, Linux, Docker, high-performance computing, reproducible workflows.

EDUCATION

Ph.D. in Computer Science **2018-2022**

University of Central Florida | Orlando, FL, USA | Advisor: Dr. Ivan Garibay

M.S. in Computer Science **2013-2015**

Technical University of Kaiserslautern | Germany | Advisor: Dr. Hans Hagen

B.Tech. in Computer Science and Engineering **2006-2010**

Jawaharlal Nehru Technological University | India